

Student Number	
Mark /	

Chemistry Assessment

Task 3 Term 2 2009

Part 1. *Theory*

General Instructions

Theory and Research

- **Reading time** – 5 minutes
- **Working time** – 110 minutes
- **Write using black or blue pen**
- **Write your Student Number at the top of this page and on page 6**
- **Board-approved calculators may be used**

A data sheet and a periodic table are provided at the back of the theory paper.

Total Marks – 46

Part A – 8 marks

- Attempt Questions 1-8

Part B – 38 marks

- Attempt Questions 9-16

Part A: Multiple Choice: 8 marks
Attempt Questions 1-8

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

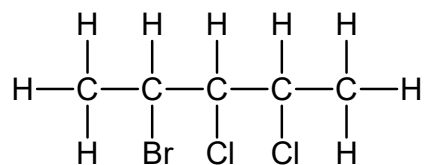
A  B  C  D 

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word correct and drawing an arrow as follows.

A  B  C  D 

▶ Mark your answers for the multiple choice questions in the multiple choice grid on page 6

1. The structural formula for a haloalkane is shown.



What is the IUPAC name of an isomer for this haloalkane?

- (A) 2-bromo-3,4-dichloropentane
- (B) 4-bromo-2,3-dichloropentane
- (C) 2-bromo-3,3-dichloropropane
- (D) 1-bromo-3,3-dichloropentane
2. Which option lists the layers of the atmosphere in the descending order.
(highest to lowest location)
- (A) thermosphere, mesosphere, stratosphere, troposphere
- (B) troposphere, mesosphere, stratosphere, thermosphere
- (C) stratosphere, mesosphere, troposphere, thermosphere
- (D) thermosphere, stratosphere, troposphere, mesosphere
- 3.. The table shows the major tropospheric air pollutants and their sources. Which of the items in the list is **incorrect**?

<i>Pollutant</i>	<i>Source</i>
(A) ozone	diffusion from the stratosphere
(B) airborne lead	lead smelters
(C) sulfur dioxide	metal smelters
(D) hydrocarbons	vehicles and factories using solvents

4. A student performed some tests to identify the cations of three nitrates, solutions *P*, *Q* and *R*. Her observations are described below :

Solution P : A white precipitate formed with sodium sulfate solution.
A brick-red colour resulted when solution *P* was sprayed into a Bunsen flame.

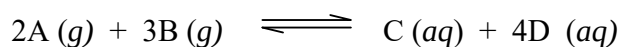
Solution Q : A yellow precipitate formed with potassium iodide solution.
No colour change was observed when solution *Q* was sprayed into a Bunsen flame.

Solution R A blue precipitate formed with sodium hydroxide solution.. A blue-green colour was observed when solution *R* was sprayed into a Bunsen flame.

What are the cations present in each of solutions *P*, *Q* and *R* ?

- (A) calcium, sodium, barium
 - (B) barium, sodium, copper
 - (C) barium, lead, copper
 - (D) calcium, lead, copper
5. Which of the following solutions can be used to determine the presence of carbonate ions in a compound?
- (A) dilute silver nitrate solution
 - (B) dilute sodium sulfate solution
 - (C) dilute nitric acid solution
 - (D) dilute barium nitrate solution

6. Consider the following reaction



Which correctly represents the equilibrium expression?

(A) $\frac{[A]^2 [B]^3}{[C] [D]^4}$

(B) $[C] [D]^4$

(C) $\frac{[C][4D]}{[2A][3B]}$

(D) $\frac{[C][D]^4}{[A]^2 [B]^3}$

7. Which of the following best describes the industrial reaction conditions for the Contact Process?

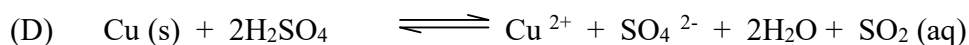
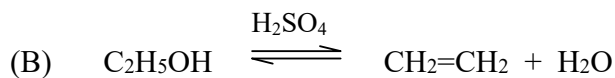
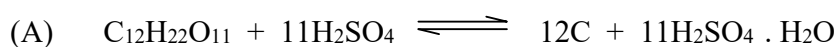
(A) 450°C V₂O₅ catalyst 2 atm

(B) 450°C V₂O₅ catalyst 200 atm

(C) 550°C Fe₃O₄ catalyst 2 atm

(D) 550°C Fe₃O₄ catalyst 200 atm

8. Which of the following shows sulfuric acid acting as an oxidizing agent?



Student Number

Part A . Answer grid for multiple choice questions

Total/ 8...

- | | | | | |
|----|-----|-----|-----|-----|
| 1. | A O | B O | C O | D O |
| 2. | A O | B O | C O | D O |
| 3. | A O | B O | C O | D O |
| 4. | A O | B O | C O | D O |
| 5. | A O | B O | C O | D O |
| 6. | A O | B O | C O | D O |
| 7. | A O | B O | C O | D O |
| 8. | A O | B O | C O | D O |

Part B: Free Response Questions

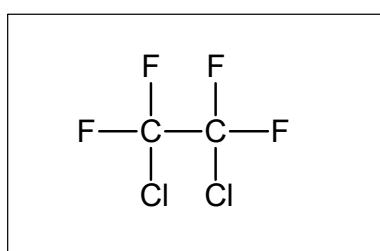
Attempt Questions 9- 16

► Show all relevant working in questions involving calculations.

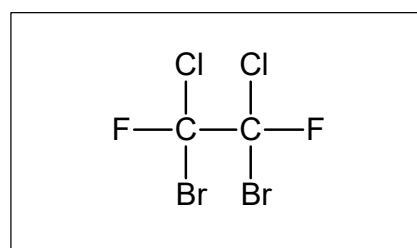
MARKS

Question 9 (3 marks)

- (a) Two molecules are shown.



Molecule 1



Molecule 2

Identify which molecule would be classified as a halon.

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- (b) Identify a possible source of halons in the atmosphere

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- (c) Describe one use of chlorofluorocarbons.

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Test continues next page

Question 10 (6 marks)

- (a) Explain with the aid of chemical equations why chlorofluorocarbons are harmful to the ozone layer. **3**

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- (b) Assess the effectiveness of the processes used to prevent the destruction of the ozone layer. **3**

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Question 11 (3 marks)

Explain the effect of ozone in the stratosphere. Include a relevant chemical equation.

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Test continues next page

Question 12 (5 marks)

Assess the role of determining dissolved oxygen and biochemical oxygen demand in ensuring the maintenance of a healthy waterway.

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Question 13(6 marks)

A student carried out a first-hand investigation to analyse the sulfate content of lawn fertilizer consisting of ammonium sulfate. The student weighed out 2.00 g of fertilizer and dissolved it in water. The student then added 50.0 mL of 0.25 molL⁻¹ barium chloride solution to the dissolved fertilizer. A white precipitate of barium sulfate formed. After filtering and drying, the barium sulfate weighed 1.8g

- (a) Calculate the percentage by mass of sulfate in the fertilizer 2

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- (b) In your practical work you carried out the analysis of sulfate in ammonium sulfate fertiliser. In the light of this work, evaluate the accuracy and reliability of the results from the above first-hand investigation. 4

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Test continues next page

Question 14 (3 marks)

Assess the impact of atomic absorption spectroscopy (AAS) on the scientific understanding of the effects of trace elements.

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Test continues next page

Question 15 (6 marks)

Sulfuric acid is the most produced industrial chemical in the world.

Complete the following table to account for the provided information about the storage and handling of sulfuric acid.

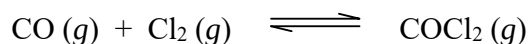
6

<i>Information</i>	<i>Explanation</i>
When diluting sulfuric acid, always add acid to water.	
Concentrated sulfuric acid can be stored in steel containers.	
Protective clothing and eyewear must be worn when using sulfuric acid.	

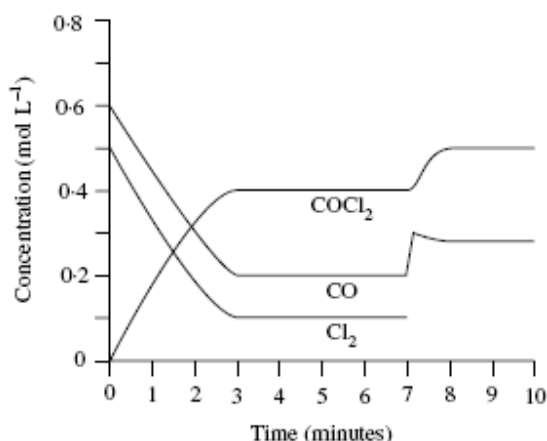
Test continues next page

Question 16 (6 marks)**MARKS**

Phosgene is produced from chlorine and carbon monoxide according to the following reaction.



When CO and Cl₂ are mixed in the presence of activated carbon, the concentration of each gas change according to the graph below.



- (a) Calculate the equilibrium constant for the reaction at 5 minutes.

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- (c) On the graph, sketch the change in concentration of chlorine after the 7 minute mark

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- (d) Explain the changes to the graph after the 7 minute mark .

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End of Theory Test



Student Number	
Mark /	

Chemistry Assessment

Task 3 Term 2 2009

Part 2: Research

General Instructions *for Theory and Research*

- Reading time – 5 minutes
- Working time – 110 minutes
- Write using black or blue pen
- Write your Student Number at the top of this page.
- Board-approved calculators may be used

Total Marks – 35

Attempt questions 1-8

Research

MARKS

Question 1 (3 marks)

Present information from your research on how you can qualitatively identify heavy metal pollutants in waterways

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Question 2 (5 marks)

(a) What is eutrophication?

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(b) Describe in detail a test to indicate the potential of a waterway for eutrophication.

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Question 3 (3 marks)

Describe the Sydney catchment in terms of catchment area and the possible sources of contamination in the catchment.

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Test continues next page

Question 4 (6 marks)

- (a) Describe how stratospheric ozone concentrations can be measured.

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- (b) On the coordinates below, draw a line graph to show the trend in mean ozone levels in the Earth's stratosphere over the past 40 years. *Note: No points need to be plotted.*

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- (c) Describe the implication of this trend.

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Question 5 (4 marks)

The concentrations of ions in substances used by society need to be monitored. Justify this statement with reference to ONE cation and ONE anion you have studied in your research

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Question 6 (5 marks)

- (a.) Identify two alternative chemicals used to replace CFCs **1**

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- (b) Evaluate the effectiveness of the use for each of the alternative chemicals identified above as replacements for CFCs **4**

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Question 7 (4 marks)

Physical and chemical processes are used to purify town water supply.
The water from a dam contains the following impurities:

- (i) dissolved gases (eg. H_2S)
- (ii) coarse pebbles and plant fragments
- (iii) fine silt making it turbid
- (iv) bacteria (e.g. *E.coli*)

Complete the table below to describe the processes and additives required to produce a safe, healthy and pleasant tasting water.

4

<i>Impurities</i>	<i>Process or additive</i>
dissolved gases	
coarse pebbles and plant fragments	
fine silt	
bacteria	

Question 8 (5 marks)

In your Industrial Chemistry Option, you researched one natural resource that is not a fossil fuel.

Name the natural resource researched . Identify the issues associated with the increased need for this natural resource and evaluate progress being made to solve the problems identified.

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End of Research Test 🔔



Student Number	
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Chemistry Assessment

Task 3 Term 2 2009

Part 1. *Theory*

General Instructions

Theory and Research

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- Working time – 110 minutes
- Write using black or blue pen
- Write your Student Number at the top of this page
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A data sheet and a periodic table are provided at the back of the paper.

Total Marks – 46

Part A – 8 marks

- Attempt Questions 1-8
- Allow about 5 minutes for this part

Part B – 38 marks

- Attempt Questions 9-16
- Allow about 45 minutes for this part

Allow about 5 minutes for this part

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 A ☐ B ☒ C ☐ D ☐

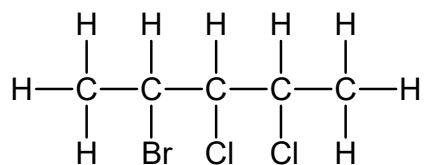
A  B  C  D 

A  B  C  D 

correct (arrow pointing to B)

page 2 of 19

1. The structural formula for a haloalkane is shown.



What is the IUPAC name **of an isomer** for this haloalkane?

- (A) 2-bromo-3,4-dichloropentane
 (B) 4-bromo-2,3-dichloropentane
 (C) 2-bromo-3,3-dichloropropane
(D) 1-bromo-3,3-dichloropentane

The question asks for the isomer!

2. Which option lists the layers of the atmosphere in the descending order.

- (A) thermosphere, mesosphere, stratosphere, troposphere**
 (B) troposphere, mesosphere, stratosphere, thermosphere
 (C) stratosphere, mesosphere, troposphere, thermosphere
 (D) thermosphere, stratosphere, troposphere, mesosphere

Outcome: H13

- 3.. The table shows the major tropospheric air pollutants and their sources. Which of the items in the list is **incorrect**?

<i>Pollutant</i>	<i>Source</i>
(A) ozone	diffusion from the stratosphere
(B) airborne lead	lead smelters
(C) sulfur dioxide	metal smelters
(D) hydrocarbons	vehicles and factories using solvents

Outcome: H4, H13

4. A student performed some tests to identify the cations of three nitrates, solutions *P*, *Q* and *R*. Her observations are described below :

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A brick-red colour resulted when solution *P* was sprayed in a Bunsen flame.

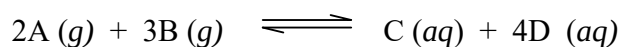
Solution Q : A yellow precipitate formed with potassium iodide solution.
No colour change was observed when solution *Q* was sprayed into the Bunsen flame.

Solution R A blue precipitate formed with sodium hydroxide solution.. A blue-green colour was observed when solution *R* was sprayed into a Bunsen flame.

What are the cations present in each of solutions *P*, *Q* and *R* ?

- (A) calcium, sodium, barium
 - (B) barium, sodium, copper
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5. Which of the following solutions can be used to determine the presence of carbonate ions in a compound?
- (A) dilute silver nitrate solution
 - (B) dilute sodium sulfate solution
 - (C) dilute nitric acid solution**
 - (D) dilute barium nitrate solution

6. Consider the following reaction



Which correctly represents the equilibrium expression?

(A) $\frac{[A]^2 [B]^3}{[C] [D]^4}$

(B) $[C] [D]^4$

(C) $\frac{[C] [4D]}{[2A] [3B]}$

(D) $\frac{[C] [D]^4}{[A]^2 [B]^3}$

7. Which of the following best describes the industrial reaction conditions for the Contact Process?

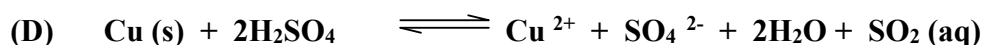
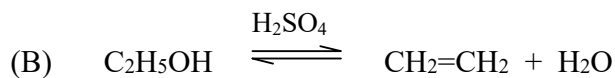
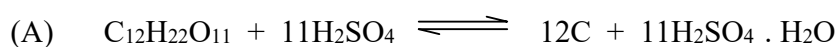
(A) **450°C V₂O₅ catalyst 2 atm**

(B) 450°C V₂O₅ catalyst 200 atm

(C) 550°C Fe₃O₄ catalyst 2 atm

(D) 550°C Fe₃O₄ catalyst 200 atm

8. Which of the following shows sulfuric acid acting as an oxidizing agent?



Part A . Answer grid for multiple choice questions**Total/8...**

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|----|-----|-----|-----|-----|
| 1. | A O | B O | C O | D ● |
| 2. | A ● | B O | C O | D O |
| 3. | A ● | B O | C O | D O |
| 4. | A O | B O | C O | D ● |
| 5. | A O | B O | C ● | D O |
| 6. | A O | B O | C O | D ● |
| 7. | A ● | B O | C O | D O |
| 8. | A O | B O | C O | D ● |

Part B Free Response Questions

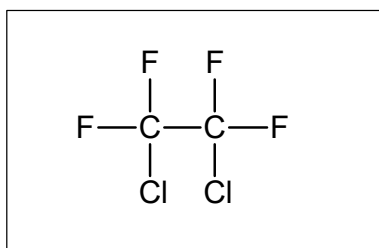
Attempt Questions 9- 16

Allow about 45 minutes for this part

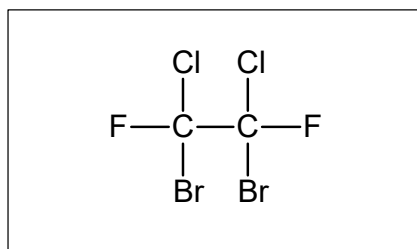
► **Show all relevant working in questions involving calculations.**

Question 9 (3 marks)

- (a) Two molecules are shown.



Molecule 1



Molecule 2

Identify which molecule would be classified as a halon. (1 mark)

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Answer

Molecule 2

- (b) Identify a possible source of halons in the atmosphere (1 mark)

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Answer:

Sources of halons include: fire extinguishers,

- (c) Describe one use of chlorofluorocarbons. (1 mark)

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Answer:

CFCs may be used as refrigerants, aerosol propellants,

Question 10 (5 marks)

- (a) Explain with the aid of chemical equations why chlorofluorocarbons are harmful to the ozone layer. (3 marks)

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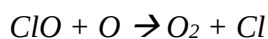
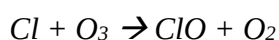
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Sample Answer



Chlorofluorocarbons may react to form chlorine atoms which react with ozone, forming oxygen and chlorine oxide. This process can continue since the chlorine atom is regenerated from chlorine oxide reacting with oxygen atoms to form oxygen molecules and chlorine.

Outcome criteria	Marks
Two chemical equations provided.	2
One chemical equation provided.	1

- (b) Assess the effectiveness of the processes used to prevent the destruction of the ozone layer. (3 marks)

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Sample Answer

Process 1: phasing out of the use of CFCs and halons via the Montreal Protocol. This has been successful since the amount of CFCs released has been decreasing since the introduction of the Montreal Protocol.

Process 2: replacement of CFCs and halons with hydrofluorocarbons which do not form chlorine atoms in the upper atmosphere. This has been successful since most processes that use CFCs and halons now use HFCs.

Outcome criteria	Marks
Describes two processes used to prevent the destruction of the ozone layer. Assesses the effectiveness of one of the processes in reducing the destruction of the ozone layer.	3
Describes two processes used to prevent the destruction of the ozone layer.	2
Describes one process used to prevent the destruction of the ozone layer.	1

Question 11 (3 marks)

Explain the effect of ozone in the stratosphere. Include a relevant chemical equation.

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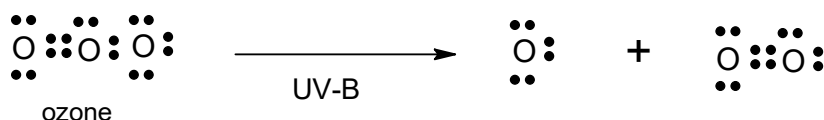
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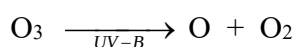
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Outcomes: H7, H13

Answer:



or



High energy UV radiation such as UV-C and UV-B are extremely harmful to living cells, causing cancer growth etc . Without ozone, life cannot exist on Earth. Oxygen filters off the UV- C, ozone, in turn, filters off UV-B resulting in its decomposition to oxygen. UV-A, the lowest energy UV, passes through the atmosphere unchanged through to the troposphere where it may be useful in small quantities in synthesising Vitamin D in the body.

Criteria	Marks
Equation showing destruction of ozone by UV-B. (Lewis structure is not required).	1
Explanation of how this destruction is useful to life on earth. (<i>filters off harmful UV light, etc</i>)	1
Effect of ozone on life in the troposphere.	1

Question 12 (5 marks)

Assess the role of determining dissolved oxygen and biochemical oxygen demand on ensuring the maintenance of a healthy waterway.

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Outcomes: H16, H13

ANS:

A healthy waterway consists of water fit as a source of drinking water (after proper treatment) and which supports diverse lifeforms such as various kinds of plants and fishes.

(D) *Dissolved oxygen is a measure of the concentration of oxygen dissolved in the water in mg/L.*

(ID) *The need for oxygen among organisms differ. The more oxygen dissolved, the more types of organism are able to survive. Low dissolved oxygen limits diversity in the waterway.*

(B) *The biochemical oxygen demand (BOD) is a measure of the requirement for oxygen of the organic materials in the water.*

(IB) *A polluted waterway will have a large amount of organic material and therefore will require more oxygen and hence higher BOD.*

(A) *BOD measurement, **therefore**, has a critical role in determining the suitability of water for household use after purification and also it is critical in determining the continuing health of the water, and the variety of organisms able to survive in the the waterway, i.e., a measure for the tendency for eutrophication.*

<i>Criteria</i>	<i>Marks</i>
statement of what dissolved oxygen is and what biochemical oxygen demand is	2
statement of the importance of dissolved oxygen and biochemical oxygen demand in assessing water quality	2
an explicit statement of the important role of DO and BOD measurement in determining the health of a waterway	1

Question 13(5 marks)

A student carried out a first-hand investigation to analyse the sulfate content of lawn fertilizer consisting of ammonium sulfate. The student weighed out 2.00 g of fertilizer and dissolved it in water. The student then added 50.0 mL of 0.25 mol L^{-1} barium chloride solution to the dissolved fertilizer. A white precipitate of barium sulfate formed. After filtering and drying the barium sulfate weighed 1.8g

- (a) Calculate the percentage by mass of sulfate in the fertilizer (2 marks)

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Sample Answer

$$\begin{aligned}n(\text{BaSO}_4) &= 1.8 / 233.37 = 7.7 \times 10^{-3} \text{ mol} \\n(\text{SO}_4^{2-}) &= n(\text{BaSO}_4) = 7.7 \times 10^{-3} \text{ mol} \\m(\text{SO}_4^{2-}) &= 7.7 \times 10^{-3} \times 96.07 = 0.74\text{g}\end{aligned} \quad (1 \text{ mark})$$

$$\% \text{mass SO}_4^{2-} = (0.74 / 2.00) \times 100 = 37\% \quad (1 \text{ mark})$$

Marking Criteria

Criteria	Mark
Correct calculation of % mass sulfate in fertilizer and correct calculation of moles of Barium sulfate and thus sulfate ions.	2
Correct calculation of moles of barium sulfate and thus moles of sulfate ions.	1

- (b) In your practical work you carried out the analysis of sulfate in ammonium sulfate fertiliser. In the light of this work, evaluate the accuracy and reliability of the results from the above first-hand investigation (4 marks)

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Sample answer

The accuracy of the results may have been affected by a number of factors including the level of solubility of the fertilizer; acid should have been added to assist the dissolving of the fertilizer. Barium sulfate is a very fine precipitate and most of it would have passed through filter paper if it had been used. The use of a sintered glass funnel would have captured a greater amount of the barium sulfate residue. The results would be more reliable if repeat trials had been conducted using the same procedure.

Marking criteria

Criteria	Mark
Evaluates the accuracy and reliability of the results of the first-hand investigation with <i>suggestions of improvements (?)</i>	4
Describes the accuracy and reliability of the results	2
Describes accuracy or reliability of the first-hand investigation	1

Question 14 (3 marks)

Assess the impact of atomic absorption spectroscopy (AAS) on the scientific understanding of the effects of trace elements.

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Sample Answer

AAS spectroscopy is used to measure the concentration of metal ions to ppm or ppb. Trace elements are needed in very small amounts by living things. Zinc, copper and iron are all trace elements found in soil and limited amounts can affect agricultural productivity. Thus AAS has allowed monitoring of levels of trace elements in the soil to levels not possible before. If levels are too low chemicals can be added to the soil to increase the amounts of the trace elements.

Marking Criteria

<i>Criteria</i>	<i>Mark</i>
Assesses the impact of AAS on the understanding of the effects of trace elements.	3
Describes the impact of AAS on the scientific understanding of the effects of trace elements	2
Describes AAS or identifies trace elements	1

Question 15 (6 marks)

Sulfuric acid is the most produced industrial chemical in the world.

Complete the following table to account for the provided information about the storage and handling of sulfuric acid.

<i>Information</i>	<i>Explanation</i>
When diluting sulfuric acid, always add acid to water.	
Concentrated sulfuric acid can be stored in steel containers.	
Protective clothing and eyewear must be worn when using sulfuric acid.	

Answer

Sulfuric acid is the most produced industrial chemical in the world.

Complete the following table to account for the provided information about the storage and handling of sulfuric acid. (6 marks)

Sample Answers

Information	Explanation
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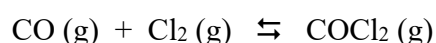
When diluting sulfuric acid, always add acid to water.	<i>The ionization of sulfuric acid is exothermic, releasing lots of heat energy. If water is added to concentrated sulfuric acid, the heat released will make the solution boil violently, splattering drops of acid. Adding acid to water helps to dissipate the heat in a larger volume.</i>
Concentrated sulfuric acid can be stored in steel containers.	<i>Concentrated sulfuric acid occurs mostly as intact molecules, thus there are hardly any hydrogen ions present to react with the steel.</i>
Protective clothing and eyewear must be worn when using sulfuric acid.	<i>Acid is corrosive and can cause burning to the skin and eyes. Wash off any acid immediately.</i>

Outcomes : H11, H8, H7, H3, H4

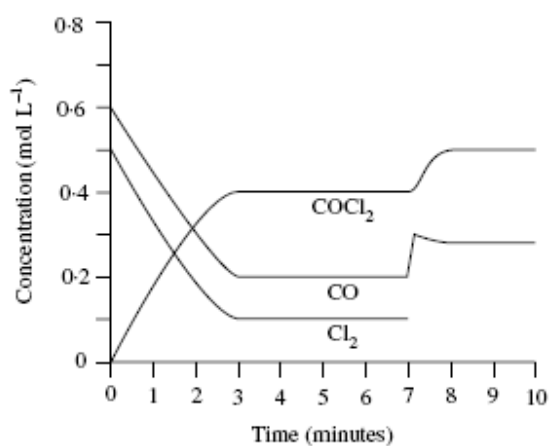
Marking Criteria	Marks
• Explains the implications of all three situations	4 -6
• Outlines the implications of all three situations	3
• Explains one situation or gives a brief outline	2
• Outlines one situation	1

Question 16 (6 marks)

Phosgene is produced from chlorine and carbon monoxide according to the following reaction.



When CO and Cl₂ are mixed in the presence of activated carbon, the concentration of each gas change according to the graph below.



- (a) Calculate the equilibrium constant for the reaction at 5 minutes. (2 marks)

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- (c) On the graph, sketch the change in concentration of chlorine after the 7 minute mark (1 mark)

- (d) Explain the changes to the graph after the 7 minute mark . (3 marks)

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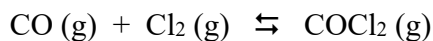
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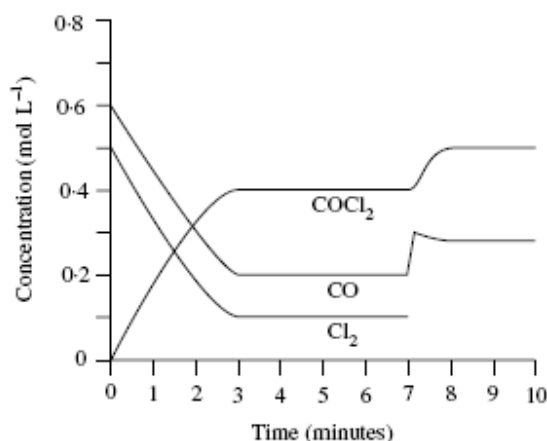
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Answer

Phosgene is produced from chlorine and carbon monoxide according to the following reaction.



When CO and Cl₂ are mixed in the presence of activated carbon, the concentration of each gas change according to the graph below.



- (a) Calculate the equilibrium constant for the reaction at 5 minutes. (2 marks)

$$K = \frac{[\text{COCl}_2]}{[\text{CO}][\text{Cl}_2]} \quad (1 \text{ mark}) \quad = \quad \frac{0.4}{0.2 \times 0.1} = 20 \quad (1 \text{ mark})$$

- (c) Sketch the change in concentration of chlorine on the graph after the 7 minute mark (1 mark)

- (d) Explain the changes to the graph after the 7 minute mark . (3 marks)

Sample Answer

At the 7 minute mark, CO was added to the system. Equilibrium shifted to oppose the change (Le Chatelier), ie to use up some CO. The concentration of CO decreased from the added amount, Cl₂ decreased and COCl₂ increased and equilibrium shifted to the right.

Marking Criteria	Marks
• Identifies added CO and explains in terms of Le Chatelier	3
• Identifies added CO and outlines equilibrium shift	2
• Identifies added CO or explains equilibrium shift	1

End of Theory Test



Student Number	
Mark /	

Chemistry Assessment

Task 3 Term 2 2009

Research Answers

General Instructions *for Theory and Research*

- Reading time – 5 minutes
- Working time – 110 minutes
- Write using black or blue pen
- Write your Student Number at the top of this page and on the response sheets on pages
- Board-approved calculators may be used

Total Marks – 35

Attempt questions 1-8

Research

MARKS

Question 1 (3 marks)

Present information from your research on how you can qualitatively identify heavy metal pollutants in waterways

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Outcomes: H11, H13

Sample Answer

Very low concentrations of heavy metal pollutants in the water can be determined using various spectrophotometric and electrochemical techniques such as anodic stripping voltammetry. The spectrophotometric technique is usually done by adding a reagent to the metal in question and allowing colour to develop. Development of a colour is an indication of the presence of the ion. Anodic stripping voltammetry is a very sensitive way of determining the identity of the elements as well as the concentration for multielement detection of pollutants.

Precipitation with sodium sulfide or other suitable reagent are also alright but not gravimetric analysis which is quantitative.

Criteria	Marks
mentions a technique to be used for analysis	1
identification by precipitation is acceptable	
explains the technique mentioned	2

Question 2 (5 marks)

(a) What is eutrophication?

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(b) Describe in detail a test to indicate the potential of a waterway for eutrophication.

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Outcomes: H4, H13, H11

ANS:

(a) *Eutrophication is the nutrient enrichment of a water body due to the presence of high levels of nitrogenous and phosphate residues resulting in the abundant growth of algae and water*

(b) *The best indicator of the tendency for eutrophication is the phosphate concentration in the water*

An algal bloom is likely if PO_4^{3-} level in a dam or lake > 0.05 ppm or > 0.1 ppm in a river or stream or a N:P ratio of greater than 10:1.

The level of phosphate can be determined by treating the sample with ammonium molybdate, reducing the resulting compound with hydroquinone or ascorbic acid to produce the highly coloured molybdenum blue. The intensity of the colour can be related to the concentration of the phosphate.

(b)

Criteria	Marks
<i>indicator of eutrophication</i>	<i>1</i>
<i>minimum level of the indicator that results in eutrophication</i>	<i>1</i>
<i>description of the technique for determination</i>	<i>2</i>

Question 3 (3 marks)

Describe the Sydney catchment in terms of catchment area and the possible sources of contamination in the catchment.

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Outcome: H4

ANS:

The Sydney catchment area consists of a large area of rivers and streams from the blue mountains to south of Mittagong. Other information that are considered:

- *extend from north of Lithgow in the upper Blue Mountains to the source of the Shoalhaven river near Cooma, from Woronora to the source of the Wollondilly River west of Crookwell*
- *65 km from city*
- *16000 km² in area*
- *Warragamba catchment is 9051 km²*
- *Lake Burragorang can contain 2,031,000 megalitres of water*

The catchment encompasses, a farm, a disused mine and bushland. These are possible sources of contamination of the water. The farm can contribute to fecal and fertiliser residue contamination of the water supply through runoff. Fecal contamination can also come from the animals living in the catchment area. (The sources of contamination must be specific to the Sydney catchment.)

Criteria	Marks
<i>description of the Sydney catchment</i>	<i>1</i>
<i>two sources of contamination in the catchment</i>	<i>2</i>

Question 4 (6 marks)

- (a) Describe how stratospheric ozone concentrations can be measured. (2 marks)

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Sample Answers

Ozone levels can be measured by spectrophotometers. These instruments may be located on the ground or in satellites and balloons. These instruments compare the intensity of the sun's light energy absorbed by ozone with the intensity of the light energy the sun does not absorb. These measurements are calibrated to give a measure of the total ozone per unit area of the Earth's surface at a particular point.

Outcome criteria	Marks
Describes how an identified instrument measures ozone levels.	2
Identifies an instrument that measures ozone levels.	1

- (b) On the coordinates below, draw a line graph to show the trend in mean ozone levels in the Earth's stratosphere over the past 40 years. Note: No points need to be plotted. (2 marks)



Sample Answer

*Graph shows a decrease in ozone over the past 40 years (1);
X axis labelled as time (year); Y-axis labelled as ozone level. (1)*

- (c) Describe the implication of this trend. (2 marks)

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Sample Answer

Depletion of ozone allows larger than normal levels of harmful radiation from the sun to reach the Earth's surface. This energy (such as UV light) may cause cancer, reduced plant growth and increase damage to chemicals such as polymers.

Outcome criteria	Marks
Identifies lower levels of ozone allowing more harmful radiation to reach the Earth's surface. Describes one potential problem associated with increased amounts of harmful radiation.	2
Identifies lower levels of ozone allowing more harmful radiation to reach the Earth's surface.	1

Question 5 (4 marks)

The concentrations of ions in substances used by society need to be monitored. Justify this statement with reference to ONE cation and ONE anion you have studied in your research

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Sample Answer

Cation –Lead is a poison; it retards brain development in young children, causes brain damage and can lead to neurological disorders. Until recently it was often used in petrol and house paints. Monitoring lead concentrations in soil near highways, in waterways and in the atmosphere in urban areas is essential to ensure that people are not exposed to harmful concentrations.

Anion : Phosphate concentration in waterways is required for normal aquatic growth but high concentrations can set off algal blooms where excessive amounts of algae can completely cover the surface of the waterway and kill fish and cause other detrimental effects. It is important to prevent algal blooms by monitoring amounts of phosphates in waterways and consumer products which can get into waterways.

Marking Criteria

<i>Criteria</i>	<i>Marks</i>
Justification of the need to monitor one specific cation and one specific anion in the research.	4
Justification of the need to monitor one either cation or anion and identification of the other cation or anion that needs to be monitored.	3
Justification of the need to monitor either one cation or anion in the research.	2
Identification of either one cation or anion that needs to be monitored in the research.	1

Question 6 (5 marks)

- (a.) Identify two alternative chemicals used to replace CFCs (1 marks)

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Sample Answer

HCFC (hydrochlorofluorocarbons) and HFC (hydrofluorocarbons)

Marking Criteria

<i>Criteria</i>	<i>Marks</i>
Identification of 2 alternative chemicals used to replace CFCs	1
Identification of one alternative chemical used to replace CFCs	0

- (b) Evaluate the effectiveness of the use for each of the alternative chemicals identified above as replacements for CFCs (4 marks)

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Sample Answer

HCFCs contain C-H bonds which are decomposed to a large extent in the troposphere so only a small proportion reach the stratosphere, thus reducing the ozone-destroying capacity of these chemicals.

HFCs contain no chlorine, only hydrogen, fluorine and carbon. The C-H bonds decompose to some extent in the troposphere, but has a zero capacity to destroy ozone. These chemicals are less efficient and more expensive than CFCs

Marking Criteria

<i>Criteria</i>	<i>Marks</i>
Evaluates the effectiveness of both alternatives identified as replacements for CFCs	4
Evaluates the effectiveness of one alternative to CFCs and describes the effectiveness of the other chemical	3
Describes the effectiveness of both alternatives chemicals to CFCs	2
Describes the effectiveness of one alternative chemical to CFCs.	1

Question 7 (4 marks)

Physical and chemical processes are used to purify town water supply.
The water from a dam contains the following impurities:

- (i) dissolved gases (eg. H_2S)
- (ii) coarse pebbles and plant fragments
- (iii) fine silt making it turbid
- (iv) bacteria (eg. *E.coli*)

Complete the table below to describe the processes and additives required to produce a safe, healthy and pleasant tasting water.

<i>Impurities</i>	<i>Process or additive</i>
dissolved gases	
coarse pebbles and plant bits	
fine silt	
bacteria	

Answer:

<i>Impurity</i>	<i>Process or additive and their action</i>
dissolved gases	<i>Aeration, water is thrown up the air, oxygen is introduced in the process oxidizing any gaseous or dissolved impurity and expelling the dissolved gases (1 mark)</i>
coarse pebbles and plant bits	<i>Water is filtered through screen and sand beds, large particles are retained by the screens (1 mark)</i>
fine silt	<i>Lime (calcium hydroxide) is added to adjust pH, then iron (III) chloride is added precipitating gelatinuous $\text{Fe}(\text{OH})_3$, precipitate (1 mark) This precipitate is capable of adsorbing fine silt. The precipitate is allowed to settle and then filtered (1 mark)</i>
bacteria	<i>Chlorine in the form of Cl_2 gas is introduced to the water to kill the bacteria</i>

Outcome: H13

<i>Marking Criteria</i>	<i>Marks</i>
Correct answer for each type of impurity	1 mark each

Question 8 (5 marks)

In your Industrial Chemistry Option, you researched one natural resource that is not a fossil fuel.

Name the natural resource researched . Identify the issues associated with the increased need for this natural resource and evaluate progress being made to solve the problems identified.

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In your Industrial Chemistry Option you studied one natural resource that is not a fossil fuel.

Identify the issues associated with the increased need for this natural resource and evaluate progress being made to solve the problems identified.

Sample Answer

Aluminium is a metal that is widely used in drink cans, window frames, saucepans, boats and aeroplanes. due to its low density, strength and durability. Usage has increased enormously as more applications have been found, population increased and standard of living improved.

It is a non renewable resource and will eventually run out.

Electrolysis is used to extract Aluminium from its ore. This process uses large amounts of electricity and carbon electrodes, thus, huge amounts of carbon dioxide are produced, contributing to the amount of greenhouse gases in the atmosphere.

Progress is being made to alleviate these problems.

Aluminium is very suitable for recycling and can be repeatedly recycled with no loss of quality. One big advantage of recycling is the energy requirement is 10% that of extraction from ore so a lot of energy is saved and green house gas emissions reduced.

Research is ongoing in improvements in the extraction process, alloys being developed to make the reserves last a little longer and the use of polymers to replace aluminium.

The progress made has been very effective in extending the life of Al reserves and reducing the energy required.

Marking Criteria	Marks
• Identified increased need • Identified uses • At least 2 issues identified • Description of progress made to solve the problems • Evaluation of progress made	5
• 4 of the above	4
• 3 of the above	3
• 2 of the above	2
• 1 of the above	1

End of Research Test 